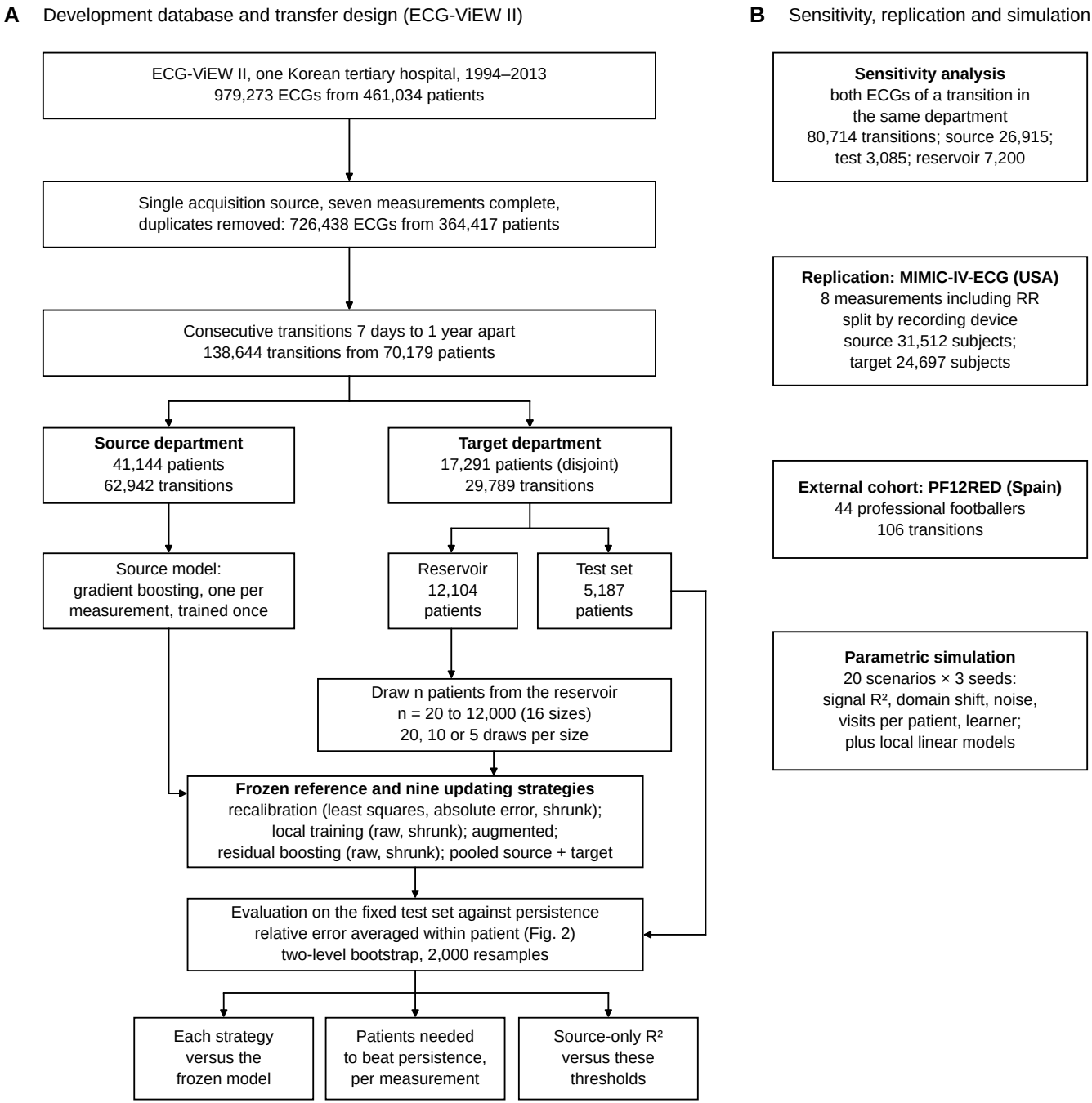
A
Development database and transfer design (ECG-ViEW II)
ECG-ViEW II, one Korean tertiary hospital, 1994–2013
979,273 ECGs from 461,034 patients
Single acquisition source, seven measurements complete,
duplicates removed: 726,438 ECGs from 364,417 patients
Consecutive transitions 7 days to 1 year apart
138,644 transitions from 70,179 patients
Source department
41,144 patients
62,942 transitions
Target department
17,291 patients (disjoint)
29,789 transitions
Source model:
gradient boosting, one per
measurement, trained once
Reservoir
12,104
patients
Test set
5,187
patients
Draw n patients from the reservoir
n = 20 to 12,000 (16 sizes)
20, 10 or 5 draws per size
Frozen reference and nine updating strategies
recalibration (least squares, absolute error, shrunk);
local training (raw, shrunk); augmented;
residual boosting (raw, shrunk); pooled source + target
Evaluation on the fixed test set against persistence
relative error averaged within patient (Fig. 2)
two-level bootstrap, 2,000 resamples
Each strategy
versus the
frozen model
Patients needed
to beat persistence,
per measurement
Source-only R²
versus these
thresholds
B
Sensitivity, replication and simulation
Sensitivity analysis
both ECGs of a transition in
the same department
80,714 transitions; source 26,915;
test 3,085; reservoir 7,200
Replication: MIMIC-IV-ECG (USA)
8 measurements including RR
split by recording device
source 31,512 subjects;
target 24,697 subjects
External cohort: PF12RED (Spain)
44 professional footballers
106 transitions
Parametric simulation
20 scenarios × 3 seeds:
signal R², domain shift, noise,
visits per patient, learner;
plus local linear models